A PROJECT TITLE

ON

Drought Severity Classification Using Multinomial Logistic Regression

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

by

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DEPARTMENT OF ______

Name of Title: Drought Severity Classification Using Multinomial Logistic Regression

1. **Abstract**:

Drought is a natural disaster that happens when there is very little rainfall for a long time. It affects farming, water supply, animals, and people's daily lives. Predicting drought early can help farmers and the government take the right actions to reduce damage.

In this project, we use **Machine Learning** to classify the severity of drought. The algorithm used is **Multinomial Logistic Regression**, which can classify data into more than two categories. The model uses weather and environmental data such as rainfall, temperature, humidity, and soil moisture to predict the drought level. The drought is classified into different categories like **No Drought, Mild Drought, Moderate Drought, and Severe Drought**.

Before training the model, the dataset is cleaned by removing missing values and preparing the data. The data is then divided into training and testing sets. After training, the model predicts the drought category for new data. The performance of the model is checked using accuracy and other evaluation methods.

This project shows that Machine Learning can help in predicting drought more quickly and accurately. The system can support farmers, researchers, and government officials in making better decisions for water management and crop planning. In the future, the model can be improved by using larger datasets and real-time weather information

2. INTRODUCTION:

Drought is a serious problem that happens when there is not enough rainfall for a long time. It can reduce crop production, cause water shortages, and affect the environment. Therefore, predicting drought early is very important.

This project uses a Machine Learning algorithm called **Multinomial Logistic Regression** to classify drought into different severity levels. The model uses weather data such as rainfall, temperature, humidity, and soil moisture to make predictions.

The main goal of this project is to build a simple and accurate model that can predict drought severity. The predicted results can help farmers and government organizations make better decisions and reduce the impact of drought.

3. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10 or Windows 11
- **Programming Language:** Python 3.x
- **IDE:** Jupyter Notebook or Visual Studio Code (VS Code)
- **Libraries:**
  - Pandas (for handling data)
  - NumPy (for mathematical calculations)
  - Scikit-learn (for Machine Learning model)
  - Matplotlib (for graphs)
  - Seaborn (for data visualization)
- **Dataset Format:** CSV (.csv)
- **Browser:** Google Chrome
- **Hardware Requirements:**
  - Processor: Intel Core i3 or above
  - RAM: Minimum 4 GB (8 GB recommended)
  - Storage: Minimum 10 GB free space

Signature of the Guide

Course Instructor